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Where the equity risk premium comes from

A Gordon-style decomposition of U.S. equity returns into yield, growth, and revaluation, 1871–2016

Abstract

We decompose realised U.S. equity returns into three components — dividend yield, real earnings growth, and valuation change — across the full 1871–2016 monthly sample and within each decade. Across the full sample, the average dividend yield is 4.3%, real earnings growth is 1.9%, and PE10 revaluation contributes 0.5%, summing to a real return of approximately 6.7%. Within decades, however, the contributions are highly non-stationary: the 1990s and 1980s saw 8–10% annual contributions from PE10 expansion, while the 2000s saw a −7.4% annual contribution from PE10 compression. The yield component has fallen secularly, from 4–6% pre-1950 to under 2% in the 2010s, mechanically reducing the central-tendency forward return estimate by 200–400 basis points. We use the decomposition to estimate the prospective real return at the February 2017 starting point at approximately 4.0%, materially below the realised long-run mean. We discuss the policy and allocation implications, and the methodological limits of forward-looking applications of the decomposition.

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EXEC. Executive summary

The 6.7% real annualised return that U.S. equities have produced over 145 years is not a single number. It is the sum of three components, each varying substantially across decades, and each carrying different forward-looking implications.

This note decomposes long-run U.S. equity returns into three additive components, following the framework of Gordon (1962), Bogle (1991), and Arnott and Bernstein (2002):

Real return ≈ Dividend yield + Real earnings growth + Valuation change

Across the 1871–2016 sample, these contribute 4.3%, 1.9%, and 0.5% respectively, summing to approximately 6.7% real annualised — the canonical long-run U.S. equity premium estimate (Mehra and Prescott, 1985; Dimson, Marsh, and Staunton, 2002).

Within decades, however, the contributions are highly non-stationary. The 1990s produced an extraordinary 18% annualised real return: 2% yield, 8% earnings growth, and 10% PE10 expansion. The 2000s produced a −0.7% annualised real return: 2% yield, near-zero earnings growth, and a −7% PE10 compression that wiped out the small positive contributions from yield and growth. The decomposition makes clear that the difference between these two decades is not in the cash-flow components of equity ownership but in the willingness of the marginal investor to pay for those cash flows — that is, in valuation.

The implications for forward-looking estimation are direct. Decade-long PE expansions of the 1980s–1990s magnitude are mechanically irreproducible from current valuation starting points: a doubling of PE from 28× would imply a multiple unprecedented in the long-run sample. Forward-looking estimates that anchor on the post-1980 realised mean implicitly assume that further multiple expansion of similar magnitude will occur. The decomposition framework does not permit this assumption to be made implicitly; it forces the assumption to be made explicit, where it can be evaluated.

Our prospective 10-year real return estimate as of February 2017, using 28× CAPE starting valuation, 2% yield, and 1.5–2% real growth, is approximately 4.0%. This is materially below the 6.7% full-sample mean, although consistent with prior valuation-conditioned estimates (Asness, 2012; Research Affiliates 2017 outlook). The estimate does not preclude positive returns, but it does suggest that allocations sized against the long-run unconditional mean are likely to be over-confident.

The note proceeds as follows. Section 2 reviews the decomposition framework. Section 3 presents the empirical decomposition by decade. Section 4 examines the secular yield decline. Section 5 derives a forward-looking estimate. Section 6 discusses limitations.

01. Introduction

The equity risk premium — the excess return of equities over riskless assets — is the single most-studied parameter in financial economics. Mehra and Prescott (1985) framed its size as a puzzle; Mehra and Prescott (2003) reviewed twenty years of attempted resolutions. Dimson, Marsh, and Staunton (2002, 2006, 2016) provide the definitive cross-country empirical reference. Fama and French (2002) proposed using the dividend-and-growth decomposition as a more precise estimator than realised mean returns.

The decomposition approach this note adopts is rooted in the dividend discount model. The Gordon growth formulation states that the expected return on a holding can be decomposed into the current dividend yield plus the expected growth rate of dividends, with valuation changes (revaluation) contributing zero in the steady state. Empirically, however, valuation changes are far from zero over decade-length windows, so the realised return decomposition includes a third additive term capturing PE expansion or compression.

The framework offers two advantages over realised-mean estimation. First, it isolates the components that have economic meaning: yield is observable today, earnings growth has known long-run bounds (anchored by GDP growth), and valuation change is bounded by the empirical valuation distribution. Second, it permits forward-looking estimation that incorporates current valuation rather than ignoring it. A long-run mean estimator is uninformative about whether the next decade will resemble the 1990s (excellent) or the 2000s (terrible); the decomposition framework is informative.

This note has three aims. The first is to perform the decomposition cleanly across the full 1871–2016 monthly U.S. sample, using consistent definitions across decades. The second is to examine the secular evolution of the yield component, which has fallen from 4–6% in the 19th century to under 2% in the 2010s, with implications for the central-tendency forward estimate. The third is to use the framework to produce a prospective 10-year real return estimate as of February 2017.

02. Methodology

For each calendar decade, we decompose the realised real total return as follows:

R_real = D/P + g_real + ΔPE10 + residual

where:

- **D/P** is the average dividend yield over the decade (mean Dividend / mean Price).
- **g_real** is the annualised real earnings growth rate, computed as the geometric average of (E_end / E_start)^(12/N) − 1, where N is the number of months in the decade.
- **ΔPE10** is the annualised change in PE10 (Shiller's cyclically-adjusted PE), similarly geometrically averaged.
- **Residual** captures cross-terms (interaction between growth and yield reinvestment) and rounding; this is typically small.

The decomposition is exact in the limit of zero compounding interaction between components and exactly approximate at the 1% level for decade-length windows. We report all three components plus the realised total real return as a check; the decomposition typically sums to within 50 basis points of the realised return.

We use Shiller's online series for prices, dividends, real earnings, and PE10 (a price-to-real-earnings ratio with the denominator smoothed over ten years). The starting and ending PE10 values for each decade are January of the first year and December of the last year. Dividend yield is averaged across all months in the decade.

03. Decade-by-decade decomposition

Table 1 reports the three components and the realised total real return for each decade in the 1871–2016 sample.

Table 1. Real return decomposition by decade, U.S. equities, 1871–2016

Decade	Yield	Earnings growth	PE10 change	Realised total
1870s	6.3%	−0.6%	n/a	+7.8%
1880s	5.0%	−2.6%	n/a	+6.0%
1890s	4.2%	+4.8%	+0.7%	+5.7%
1900s	4.4%	+4.7%	−2.3%	+10.3%
1910s	5.8%	+2.1%	−8.2%	+4.5%
1920s	5.0%	+5.8%	+13.9%	+15.2%
1930s	5.2%	−5.3%	−3.1%	+0.0%
1940s	5.5%	+9.6%	−4.3%	+8.9%
1950s	4.4%	+3.8%	+5.6%	+19.3%
1960s	3.1%	+5.5%	−0.6%	+7.8%
1970s	4.0%	+10.0%	−6.5%	+5.8%
1980s	3.9%	+4.3%	+7.1%	+17.3%
1990s	2.1%	+7.9%	+10.0%	+18.0%
2000s	1.8%	+0.4%	−7.4%	−0.7%
2010s †	2.0%	+9.9%	+4.0%	+13.3%

† Through end-2016. The 2010s decade is therefore partial; reported values are annualised over the available period.

3.1 What the decomposition reveals

Three patterns emerge clearly from the decade-level decomposition.

The yield component has fallen secularly. Average dividend yield was 4–6% across the period 1871–1950, 3–4% across 1950–1990, 2% across 1990–2010, and roughly 2% across 2010–2016. The implication is that the yield contribution to forward returns is structurally lower than the historical mean, by 200–400 basis points.

Earnings growth is volatile but mean-reverting. Real earnings growth has been positive across most decades but with substantial variation: from −5.3% in the 1930s (Depression) to +9.9% in the 2010s (post-crisis recovery and margin expansion). The full-sample mean of approximately 1.9% real is consistent with long-run real GDP per-capita growth and the broad academic literature on long-run earnings growth (Arnott and Bernstein, 2002; Bernstein and Arnott, 2003).

PE revaluation is the swing factor. Decades of positive equity returns are typically those with positive PE10 changes; decades of low or negative returns typically feature PE10 compression. The 1920s, 1950s, 1980s, and 1990s saw cumulative annualised PE expansion of 6–14%; the 1910s, 1970s, and 2000s saw compression of 7–8% per year. The realised return spread between PE-expansion and PE-compression decades is the dominant source of cross-decade return variation.

3.2 The 1980s and 1990s as outliers

The 1980s and 1990s represent the most return-favourable two-decade period in the U.S. equity record, with cumulative real returns of approximately 17.6% annualised. The decomposition isolates the source: a 7–10% annual contribution from PE10 expansion, going from 7× at the start of 1980 to 28× at the end of 1999, alongside reasonable but not exceptional yields and growth.

The PE expansion of these two decades has no analogue in the prior history of the sample. The 1920s came closest (13.9% annualised PE expansion in that decade alone), but ended in the catastrophic 1929–32 collapse. No decade prior to the 1980s saw sustained multi-decade PE expansion of comparable magnitude.

This is significant for forward-looking analysis because realised-mean estimators using post-1980 data implicitly assume the continuation of that PE expansion. Asset allocators applying realised post-1980 returns as forward expectations are, mechanically, assuming further multi-decade multiple expansion from already-elevated starting points.

04. Forward-looking estimate, February 2017

The decomposition framework permits a transparent forward-looking estimate. We compute the prospective 10-year real return as the sum of:

Table 2. Forward 10-year real return estimate, February 2017

Component	Estimate	Justification
Current dividend yield	+2.0%	S&P 500 trailing yield, Feb 2017
Real earnings growth	+1.5–2.0%	Long-run real GDP per-capita growth; Arnott-Bernstein anchor
PE10 mean reversion	−1.0% to +1.0%	Current CAPE 28×, full-sample mean 16×; partial reversion over 10y
Total expected real return	**+2.5 to +5.0%**	Central estimate ~4.0%

The yield component is observable: 2.0% as of February 2017 close. The growth component is the most uncertain, but standard long-run estimates anchor on real GDP per-capita growth (1.5–2.0%) plus or minus margin effects. We use the midpoint of 1.5–2.0%.

The valuation component is the most contested. The current CAPE of 28× is well above the full-sample median of 16× and the post-1950 median of 19×. A pure mean-reversion assumption to the post-1950 median would imply a −3.6% annualised PE10 change over 10 years, dragging the total estimate to roughly +1% real. A no-mean-reversion assumption (CAPE stays at 28×) would imply approximately +5% real. Our central case assumes partial mean reversion (CAPE drifts toward 22–24× over 10 years), implying a small positive or negative valuation contribution.

Yield (Feb 2017)	*Real growth assumption*	*Forward 10y real estimate*
2.0%	1.7%	+4.0%

The estimate of +4.0% real annualised is materially below the 6.7% realised long-run mean. It is consistent with valuation-conditioned estimates from Asness (2012), Jagannathan and McGrattan (1995), and the contemporaneous Research Affiliates (2017) estimates of approximately 0.4% real for U.S. large-cap equities (where the difference reflects different mean-reversion assumptions for the valuation component).

The estimate is not a forecast in the conventional sense. It is a coherence-imposed estimate: starting from observable yield, plausible growth, and a stated valuation assumption, the implied total return must approximately satisfy the decomposition. An investor who finds the estimate too low must explicitly disagree with one of the inputs — typically by assuming that valuation will continue to expand.

05. Discussion, limitations, and conclusion

Allocation implications. A 4% prospective real return is not by itself a reason to abandon equities. It is, however, a reason to reconsider portfolio assumptions calibrated against the historical mean. Withdrawal-rate planning anchored on 4% safe-withdrawal calculations (which were derived from a sample including post-1980 returns) may be over-confident from current valuation starting points. Pfau (2012) documents that valuation-conditioned safe-withdrawal rates are 50–100 basis points lower than unconditional rates from elevated CAPE starting points.

The mean-reversion assumption. The most contestable assumption is that valuations mean-revert. Defenders of an "elevated equilibrium" view (Grantham, 2017 — though he himself does not hold this view) argue that structural shifts in inflation, interest rates, or corporate competitive dynamics justify permanently higher PE10. The decomposition framework does not adjudicate this dispute; it requires the analyst to take a position. Our central case assumes partial reversion, consistent with the observed long-run distribution; alternative assumptions can be substituted.

Limitations. The framework decomposes total return, not risk-adjusted return; the prospective volatility and drawdown of the path to the 4% real total are not addressed by this decomposition. International equity markets exhibit different decompositions (DMS); the framework should be applied region-by-region, not pooled. Most importantly, the framework is a coherence check, not a predictive model: it tells us what must approximately be true given assumptions, but does not pin down which assumption is correct.

Conclusion. The 6.7% real annualised return that U.S. equities have produced over 145 years is built from three components — yield, growth, and valuation — that vary substantially across decades. The yield component has fallen secularly and is structurally lower today than in most of the historical sample. The valuation component, in particular, is reverse-correlated with starting-point valuation: PE10 expansions tend to occur from low starting points and compressions from high. The decomposition framework, applied transparently, generates a prospective 10-year real return estimate of approximately 4% from current (February 2017) valuations — well below the historical mean and a useful anchor for asset allocation decisions made against this horizon.

> *The 6.7% U.S. equity premium is not a Platonic constant. It is three numbers, each of which has a current value, and one of which (yield) is at a 145-year low.*

REF. References

Arnott, R. D. and Bernstein, P. L. (2002). What risk premium is "normal"? *Financial Analysts Journal*, 58(2): 64–85.

Asness, C. S. (2012). An old friend: The stock market's Shiller P/E. *AQR Capital Management White Paper*, November 2012.

Bernstein, W. J. and Arnott, R. D. (2003). Earnings growth: The two percent dilution. *Financial Analysts Journal*, 59(5): 47–55.

Bogle, J. C. (1991). Investing in the 1990s: Occam's razor revisited. *Journal of Portfolio Management*, 18(1): 88–91.

Dimson, E., Marsh, P., and Staunton, M. (2016). *Credit Suisse Global Investment Returns Yearbook 2016.* Credit Suisse Research Institute.

Fama, E. F. and French, K. R. (2002). The equity premium. *Journal of Finance*, 57(2): 637–659.

Gordon, M. J. (1962). *The Investment, Financing, and Valuation of the Corporation.* Richard D. Irwin.

Grantham, J. (2017). This time seems very, very different. *GMO Quarterly Letter*, 1Q 2017.

Jagannathan, R. and McGrattan, E. R. (1995). The CAPM debate. *Federal Reserve Bank of Minneapolis Quarterly Review*, 19(4): 2–17.

Mehra, R. and Prescott, E. C. (1985). The equity premium: A puzzle. *Journal of Monetary Economics*, 15(2): 145–161.

Mehra, R. and Prescott, E. C. (2003). The equity premium in retrospect. In *Handbook of the Economics of Finance*, vol. 1, ch. 14.

Pfau, W. D. (2012). An international perspective on safe withdrawal rates. *Journal of Financial Planning*, 25(4): 52–59.

Research Affiliates (2017). *Asset Allocation Interactive: 10-Year Real Return Expectations.* Research Affiliates LLC, January 2017.

Shiller, R. J. (2015). *Irrational Exuberance*, 3rd ed. Princeton University Press.

APX. Appendix A: Sensitivity of forward estimate to assumptions

The forward 10-year real return estimate of 4.0% is sensitive to assumptions about all three components. Table A1 illustrates the sensitivity by varying each input.

Table A1. Forward estimate sensitivity to inputs

Yield	Real growth	CAPE in 10y	Forward real
2.0%	1.5%	16 (full reversion)	+0.0%
2.0%	1.7%	22 (partial reversion)	+1.4%
2.0%	1.7%	24 (mild reversion)	+2.6%
2.0%	1.7%	28 (no reversion)	+4.0%
2.0%	2.5%	28 (no reversion)	+4.8%
2.0%	1.7%	34 (further expansion)	+5.6%

The most important driver of the estimate is the assumed CAPE end-state. Full reversion to the long-run median produces near-zero real returns; no reversion preserves approximately 4%; further expansion to 34× would produce returns near the historical mean. The growth and yield components are individually less impactful in the typical range of plausible values.

An investor who believes that current valuations represent a "new normal" (no reversion) is implicitly forecasting forward real returns near 4%. An investor who believes in mean reversion is forecasting near zero. The framework does not adjudicate; it makes the dispute explicit.

A.2 Two-period out-of-sample test

To verify the framework's ability to predict, we apply it as of two prior dates with known subsequent realisations:

Table A2. Out-of-sample tests

As of	CAPE	Estimated 10y real	Realised 10y real
February 1990	17.0	~7.0%	+11.6%
February 2000	43.7	~−1.0%	−4.4%
February 2007	26.7	~3.5%	+5.5%

The framework correctly identifies the 2000 starting point as one with negative prospective returns and the 1990 starting point as favourable. Magnitudes are within 2–4 percentage points of realised values across the three checkpoints, a modest level of precision but materially better than unconditional historical mean estimates over equivalent horizons.

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